Curriculum Vitae — Andriy V. Koval

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Professional Summary

Senior Informatician and applied researcher with 10+ years of experience in data architecture, statistical modeling, and reproducible analytics across government, health services, and academia. At the Government of Alberta, serve as lead methodologist and data scientist for ministry efforts to evaluate the impact of social services, including Income Support and Career and Employment Information Services. Apply quasi-experimental research designs and reproducible research practices to connect administrative data with evidence on program outcomes.

Recent public work connects data engineering with survey research and time-series forecasting, making methods, validation, and results available for inspection. PhD in Quantitative Methods (Vanderbilt University); former tenure-track professor at the University of Central Florida. Developed reusable project infrastructure adopted across 32 repositories and delivered 15+ analyst training sessions. Author or co-author of 12+ peer-reviewed publications.

Featured Public Projects

Absent from Chronic: chronic conditions and work absenteeism. Develop reproducible analytical workflows for a collaborative study of 63,843 employed Canadian Community Health Survey respondents. The published analysis site connects survey-variable definitions, automated data validation, missingness and outcome diagnostics, and LASSO variable selection to the research proposal. Demonstrates traceability from raw survey files to model interpretation; research in progress.

Caseload Forecast Demo: Alberta Income Support. Build an end-to-end forecasting demonstration using 246 months of public data. Compare seasonal ARIMA with a seasonal naive baseline in a 24-month holdout backtest; publish 24-month projections with 80% and 95% prediction intervals. Versioned training-data identifiers connect forecasts to their inputs, while Parquet artifacts support portability. Public learning sandbox, separate from the internal production system; cloud migration is planned.

Education

Dates	Degree
September 2008 – May 2014	**Doctor of Philosophy, Quantitative Methods**, Peabody College of Education and Human Development, Vanderbilt University, Nashville, TN. Dissertation: *A Graphical System for Longitudinal Modeling using Dynamic Documents: Application to NLSY97 Religiosity Data*.
August 2006 – May 2008	**Master of Arts, Quantitative Psychology**, Middle Tennessee State University, Murfreesboro, TN. Thesis: *Comparative analysis of modern differential item functioning (DIF) methods through Monte Carlo simulation*.
August 2001 – May 2005	**Bachelor of Science, Journalism**, Summa Cum Laude, with Distinction from Honors College, College of Mass Communication, Middle Tennessee State University, Murfreesboro, TN. Thesis: *Influence of Brand Advertising on the Nature of Interpersonal Communication in Ukrainian Society*.

Professional Experience

Senior Informatician

Assisted Living and Social Services (formerly Seniors, Community, and Social Services)
Strategic Data, Analytics and Business Services · Research and Open Data
May 2023 – Present

- Design and maintain the **Research Database (RDB)** — a reproducible SQL Server-based analytics framework connecting 15+ raw administrative data tables through standardized ETL patterns (Ferry for staging, Ellis for transformation) to a research warehouse supporting 18+ analysis notebooks
- Establish and enforce **data governance standards**: quality assurance protocols, reproducibility requirements, scripting standards, version control practices, and automated documentation (CACHE manifests) across all team projects
- Led the original **FIESTA (Financial Impact of Employment Services and Training in Alberta)** net impact analysis from study design through reporting: assembled approximately 88,000 Income Support clients and linked approximately 75,000 to tax records; assessed linkage bias and estimated service effects using propensity-score weighting and a doubly robust approach. Findings informed employment-service funding decisions
- Built the versioned **FIESTA 2.0** preparation pipeline covering 734,000 individuals and 8.6M+ records; provide analysis-ready data, reproducibility documentation, and methodology advice. Alberta's Office of Statistical Information leads linkage and model estimation for this phase
- Develop **caseload forecasting** architecture from ingestion through reporting; public open-data demonstration includes seasonal ARIMA modeling, baseline comparison, holdout validation, prediction intervals, and traceable training-data versions
- Contribute reproducible architecture, templates, and AI-assisted development to **Work Outcomes Reporting Project (WORP)** modernization with the survey manager and lead statistician, in workflows involving response verification, nonresponse weighting, and reporting alerts
- Collaborate on **Income Support open-data report preparation** from artifacts approved for release; automate analysis, visualization, and report generation, not release approval
- Integrate colleagues' **Employment Readiness Assessment (ERA)** reporting scripts into a documented, reproducible framework supporting weekly processing and reports every two weeks; retain the classification and reporting logic developed by colleagues
- Standardize **CEIS impact assessment** methodology (event → interval → intervention → analysis) at three granularity levels
- Adapt and extend Will Beasley's RAnalysisSkeleton into the **Quick-Start Template**, a shared framework used across 32 branch repositories for project organization, version control, and documentation. Since 2023, maintain the template and expand it into an agent harness for data engineering, modelling, and frontend publishing; the adoption count describes the framework, not every later AI feature
- Lead **capacity building** initiatives: 15+ Knowledge Onboarding Guide sessions; ~8 staff onboarded to R, GitHub, and AI-augmented workflows (Copilot, custom personas)
- Coordinate monthly data integration refreshes across multiple source systems, monitor pipeline integrity, and troubleshoot production issues
- Participate in cross-ministry networks: Data Analytics Network (DAN), Jobs, Economy, Trade and Immigration (JETI), Health Analytics collaboratives
- Support external research partnerships (University of Calgary, AI case notes analysis)

Data Analyst

Alberta Community and Social Services
Strategic Data & Analytics
June 2021 – April 2023

- Designed and implemented the initial Research Database framework for provisioning and analyzing administrative data with emphasis on reproducibility, scalability, and auditability
- Developed reproducible analytic pipelines for cross-ministerial research initiatives using R and SQL Server
- Provided methodology consultation for research design, statistical analysis, and data visualization
- Initiated capacity building sessions for branch analysts in R programming and GitHub-based version control

- Addressed routine and ad-hoc information requests from program areas and senior leadership

Assistant Professor (Tenure Track)

University of Central Florida, College of Community Innovation and Education
Department of Health Management and Informatics
December 2018 – June 2021

- Developed and taught graduate courses: **Health Care Database Management**, Health Care Analytics, Epidemiology, Analytics and Quality Management
- Supervised postdoctoral, graduate, and undergraduate researchers in applied research projects
- Led funded research with Veterans Affairs on therapy adherence during COVID-19 social distancing measures
- Advised Health Foundation of South Florida on survey design, psychometric monitoring, and analytic pipeline development for substance-use recovery programs
- Analyzed youth suicide trends for Florida Linking Individuals Needed Care project to inform state suicide prevention policy
- Collaborated with Homeless Services Network of Central Florida on data visualization platform for HUD system performance metrics

Health System Impact Fellow (CIHR)

BC Centre for Disease Control / University of British Columbia
BC Observatory for Population and Public Health
August 2017 – December 2018

- Won competitive national fellowship from Canadian Institutes of Health Research for health system analytics
- Designed and implemented a decision-support system for small cell suppression in epidemiological reporting
- Developed competencies in project management, knowledge translation, and change management through CIHR training program
- Coordinated with regional epidemiologists across British Columbia for systems implementation

Postdoctoral Research Appointment

One full-time appointment with shared supervision and overlapping affiliations:

University of Victoria, Institute of Aging & Lifelong Health
Integrative Analysis of Longitudinal Studies of Aging (IALSA) Network
August 2014 – August 2017

Vancouver Island Health Authority, Applied Clinical Research Unit
June 2015 – August 2017

University of Victoria / IALSA

- Led development of software tools for coordinating multi-site statistical analyses across research teams in six countries (Canada, United States, United Kingdom, Sweden, Germany, Netherlands)
- Organized and conducted training seminars, hackathons, and synchronous coding events for diverse interdisciplinary participants
- Founded **Data Science Studio (DSS)** — an interdisciplinary learning collaborative for reproducible research, housed at University of Victoria

Vancouver Island Health Authority

- Designed platform for management and analysis of encounter data for the health authority
- Advised on data governance, data disclosure processes, and implementation of a Controlled Data Disclosure Environment
- Contributed to de-identification methodology for high-dimensional transactional health datasets

Technical Skills

Category	Tools and Methods
Database and Architecture	SQL Server, SQLite, Parquet, relational database design, data modeling, ETL pipeline design (Ferry/Ellis patterns), data warehousing, versioned data lineage
Programming and Analysis	R (primary: tidyverse, ggplot2, Quarto, rmarkdown); SQL (proficient); Python (familiar); SAS, SPSS, Mplus, STATA
Statistical Methods	Net impact analysis, propensity score methods, quasi-experimental design, longitudinal and multilevel modeling, survey methodology, LASSO variable selection, seasonal ARIMA forecasting, holdout validation, prediction intervals
Reporting and Visualization	ggplot2, Quarto, R Markdown, interactive dashboards, Power BI, data storytelling, Edward Tufte visualization principles
Data Governance	Quality assurance standards, reproducibility protocols, data documentation (automated manifests), FOIP awareness, data disclosure methodology, de-identification
DevOps and Infrastructure	Git, GitHub, VS Code, RStudio, CI/CD workflows, AI-augmented development (Copilot, custom personas)
AI and Automation	AI-augmented research analytics, persona-based context systems, orchestration frameworks, Alberta AI Academy (Levels 1–3)

Key Projects (2021–Present)

Project	Description	Scale
Absent from Chronic	Collaborative survey research on chronic conditions and work absenteeism; validated data pipeline, predictor diagnostics, LASSO selection, and linked analytical reports	63,843 respondents; pooled CCHS cycles 2010-2011 and 2013-2014
Caseload Forecast Demo	Public Income Support forecasting sandbox; seasonal ARIMA and baseline backtesting, prediction intervals, versioned data lineage	246 monthly observations; 24-month horizon
Research Database (RDB)	Reproducible SQL Server-based analytics framework; Ferry/Ellis ETL patterns; staging and warehouse layers; automated manifests	15+ source tables, 18+ notebooks
Original FIESTA	Led study design, cohort assembly, linkage, propensity-score weighting, estimation, and reporting; findings informed funding decisions	~88,000 clients assembled; ~75,000 linked
FIESTA 2.0	Built versioned preparation pipeline and provide methodology advice; Office of Statistical Information leads linkage and estimation	734K individuals, 8.6M+ records
Quick-Start Template And Agent Harness	Adapted and extended RAnalysisSkeleton; shared project framework plus later agent-assisted engineering, modelling, and publishing	Framework used across 32 repositories
WORP Modernization	Shared survey-reporting modernization; contribute reproducible architecture, templates, and AI-assisted development	Monthly reporting workflow
Open-Data Report Preparation	Collaborated on automated IS caseload analysis, visualization, and report generation from release-approved artifacts	Preparation, not release approval
ERA Reporting	Integrated colleagues' scripts into a repeatable framework; retained their classification logic	Weekly processing; fortnightly reports
CEIS Impact Assessment	Standardized net impact process at three granularity levels	Methodology framework
Senior Forecasting	ASB forecasting using StatCan population projections; onboarded into template paradigm	Production system

Project	Description	Scale
Service System Simulation	Simulated 7-table ontology preserving structural properties of administrative data; privacy-safe external collaboration	Open-source
AIM 2025 Sandbox	AI Maximalists initiative; AI-enhanced research analytics with persona-based context system and orchestration	Capacity building
Books of Ukraine	Cross-domain data pipeline (Ferry/Ellis) for Ukrainian publishing data; multilingual, multi-contributor research collaboration	4-stage pipeline, SQLite

Selected Publications

- **Koval, A.**, Beasley, W. H., Hararuk, O., & Rodgers, J. L. (2023). Social Contagion and General Diffusion Models of Adolescent Religious Transitions: A Tutorial, and EMOSA Applications. *Journal of Research on Adolescence, 33*(1), 318–343.
- Unruh, L., Allin, S., Marchildon, G., Burke, S., Barry, S., Siersbaek, R., **Koval, A.**, & Williams, G. A. (2022). A comparison of 2020 health policy responses to the COVID-19 pandemic in Canada, Ireland, the United Kingdom and the United States of America. *Health Policy, 126*(5), 427–437.
- Andraka-Christou, B., Nguyen, T., Harris, S., Atkins, D. N., Totaram, R., Golan, O., **Koval, A.**, & Madeira, J. (2022). Harm reduction policy support among students at two US universities. *Substance Use & Misuse, 57*(8), 1185–1195.
- Kulkarni, S. A., Gurpur, V., King, C., & **Koval, A.** (2021). Impact of Gaussian Noise for Optimized Support Vector Machine Algorithm Applied to Medicare Payment on Raspberry Pi. *Informatica, 45*(4).
- Embrett, M., Liu, R. H., Aubrecht, K., **Koval, A.**, & Lai, J. (2021; published online 2020). Thinking Together, Working Apart: Leveraging a Community of Practice to Facilitate Productive and Meaningful Remote Collaboration. *International Journal of Health Policy and Management*.
- Chishtie, J. A., Babineau, J., Bielska, I. A., Cepoiu-Martin, M., Irvine, M., **Koval, A.**, … & Jaglal, S. (2019). Visual analytic tools and techniques in population health and health services research: protocol for a scoping review. *JMIR Research Protocols, 8*(10), e14019.
- Blanchette, M-A., Cheng, I., Saari, M., Ghandour, K., Aubrecht, K., Haw, J., Bailey, C., **Koval, A.**, & Sim, M. (2019). Making Contributions and Defining Success: An eDelphi Study of the Inaugural Cohort of CIHR Health System Impact Fellows. *Healthcare Policy, 15*(SP), 49–60.
- Duggan, E. C., Piccinin, A. M., Clouston, S., **Koval, A. V.**, Robitaille, A., … & Hofer, S. M. (2019). A multi-study coordinated meta-analysis of pulmonary function and cognition in aging. *The Journals of Gerontology: Series A, 74*(11), 1793–1804.
- Zammit, A. R., Piccinin, A. M., Duggan, E., **Koval, A.**, Hofer, S. M. (2019). A coordinated multi-study analysis of the longitudinal association between handgrip strength and cognitive function in older adults. *The Journals of Gerontology: Series B*.
- Moselle, K. A., Robertson, S., & **Koval, A.** (2019). "Real-World" De-Identification of High-Dimensional Transactional Health Datasets. *Studies in Health Technology and Informatics, 257*, 319–324.
- Robitaille, A., van den Hout, A., Machado, R. J., Bennett, D. A., Čukić, I., Deary, I. J., … & **Koval, A. V.** (2018). Transitions across cognitive states and death among older adults in relation to education: A multistate survival model. *Alzheimer's & Dementia, 14*(4), 462–472.
- Kelly, A., Calamia, M., **Koval, A.**, Muniz-Terrera, G., Piccinin, A., Clouston, S., … & Hofer, S. (2016). Independent and interactive impacts of hypertension and diabetes mellitus on verbal memory: A coordinated analysis. *Psychology and Aging, 31*(3), 262–273.
- **Koval, A. V.**, Beasley, W. H. (2014). Abstract: Exploring Crawford-Ferguson Family of Rotations through an Interactive Visual System. *Multivariate Behavioral Research, 49*, 288–289.
- Rodgers, J. L. & **Koval, A.** (2012). EMOSA models as developmental methodology. In Laursen, B., Little, T. & Card, N. (Eds.), *Handbook of Developmental Research Methodology* (pp. 701–724). New York: Guilford Press.

Selected Presentations (70+ total)

- **Koval, A.** (2020). Visualizing the Pandemic: Comparing Trajectories of COVID-19 across Countries. Featured speaker, VADA Program Summer School. (link)
- **Koval, A.**, Smolina, K., & Leamon, A. (2020). Using Reproducible Data Visualizations to Augment Decision-Making During Suppression of Small Counts. *International Journal of Population Data Science, 5*(5). IPDLN Conference. (link)
- **Koval, A.** (2019). Managing Data Analysis in RStudio using Project-Oriented Workflow. Invited workshop, CIHR Health System Impact Fellowship retreat, Toronto. (repo)
- **Koval, A.** (2019). Techniques for Reproducible Visualization in R: Graph as a Sequence of Three Functions. Featured speaker, VADA Program Summer School, Winnipeg. (repo)
- **Koval, A.** (2019). Using GitHub for Data Science: Version Control, Project Management, Promotion. Featured speaker, VADA Program Summer School, Winnipeg. (repo)
- **Koval, A.** (2019). Implementing Reproducible Visualizations: Application of coloring book technique in a reproducible ggplot2 system. Invited talk, UCF Department of Statistics and Data Science.
- Moselle, K. A., Robertson, S., & **Koval, A.** (2019). "Real-World" De-Identification of High-Dimensional Transactional Health Datasets. Paper presented at the 20th Annual Privacy and Security Conference, Victoria, BC.
- **Koval, A.**, Moselle, K. (2018). Clinical Context Coding Scheme: Describing Utilization of Services of Island Health between 2007–2017. **Best Data Visualization Award.** IPDLN Conference, Banff, AB. (link)
- **Koval, A.**, Moselle, K. (2018). Blueprints for Learning: How does Healthcare System advance the science of its data? CHSPR Conference, Vancouver, BC. (link)
- **Koval, A.** (2018). When notebooks are not enough. Invited talk, Matrix Institute for Applied Data Science, University of Victoria.
- **Koval, A.**, Piccinin, A., Hofer, S., Muniz, G., Beasley, W. H. (2015). A Collaborative Modeling Framework for Multi-Study Coordinated Analyses. Oral presentation, Wellcome Trust Conference, Cambridge, UK.
- **Koval, A.**, Beasley, W. H. (2017). Scientific Collaboration with GitHub. Invited speaker, Hawaii Institute of Marine Biology, Gates Coral Lab.

Collaborative Workshops and Hackathons

- *Linked Data Innovation Challenge*. **Hackathon** at the 2018 International Population Data Linkage Conference, Banff AB. Best Data Visualization award for dynamic visualization of logistic regression with the coloring book technique. (repo)
- *Probabilistic Programming for Advanced Machine Learning* **summer school** hosted by DARPA, Arlington VA, July 2017. Project demonstrated BayesDB application in simulating realistic EHR trained by transactional records. (repo)
- *Assessing the impact of different harmonization procedures*. Maelstrom Harmonization **workshop**, April 2016, Groningen, Netherlands. Role: developer of a workflow for harmonizing 5 longitudinal studies. (repo)
- *Multi-study and multivariate evaluation of healthy life expectancy*. IALSA **workshop** on multistate modeling, April 2016, Amsterdam, Netherlands. Role: modeler and repository developer. (repo)
- *Coordinated Analysis with replication using data from nine longitudinal studies*. IALSA **workshop**, February 2015, Portland, OR. Role: developing software tools for parallel analysis in Mplus, R, and GitHub. (repo)

Teaching Experience

Course	Institution	Level
Health Care Database Management (repo)	University of Central Florida	Graduate

Course	Institution	Level
Health Care Analytics (repo)	University of Central Florida	Graduate
Epidemiology, Analytics, and Quality Management	University of Central Florida	Graduate
Introduction to Statistical Modeling (repo)	University of Victoria	Graduate

Supervision and Mentoring

- Gabriel J. Merrin — Postdoctoral Fellow (UCF, 2019)
- Johd Pannu — Graduate Student (UCF, 2019)
- William Belanger — Graduate Student (UCF, 2020)
- Aysha Khan — Undergraduate Student (UCF, 2020)

Professional Development

- Alberta AI Academy — Levels 1, 2, and 3 (September 2025–February 2026)
- Light-touch Project Management, CIHR Health System Impact Fellowship Training, Toronto, November 2017
- Probabilistic Programming for Advancing Machine Learning, Summer School, Arlington VA, July 2017
- Advanced R Programming & R Package Development, Workshops by RStudio and Hadley Wickham, Washington DC, January 2012
- Presenting Data and Information, Workshop by Edward Tufte, San Francisco CA, May 2012
- National Longitudinal Survey Summer Workshops, Columbus OH, May 2009

Awards and Honors

- Best Data Visualization Award, International Population Data Linkage Conference, Banff AB, 2018
- Health System Impact Fellowship, Canadian Institutes of Health Research, 2017
- Travel Award, Society of Multivariate Experimental Psychology, 2013
- Outstanding Graduate Student in Quantitative Psychology, MTSU, 2008
- Summa Cum Laude, with Distinction from Honors College, MTSU, 2005

Service

- Advisory Committee Member, Visual and Automated Disease Analytics (VADA) Graduate Program, University of Manitoba
- Participant, Data Analytics Network (DAN), Government of Alberta
- Participant, Jobs, Economy, Trade and Immigration (JETI) Analytics, Government of Alberta
- Participant, Health Analytics Collaboratives, Government of Alberta

References

Available upon request.